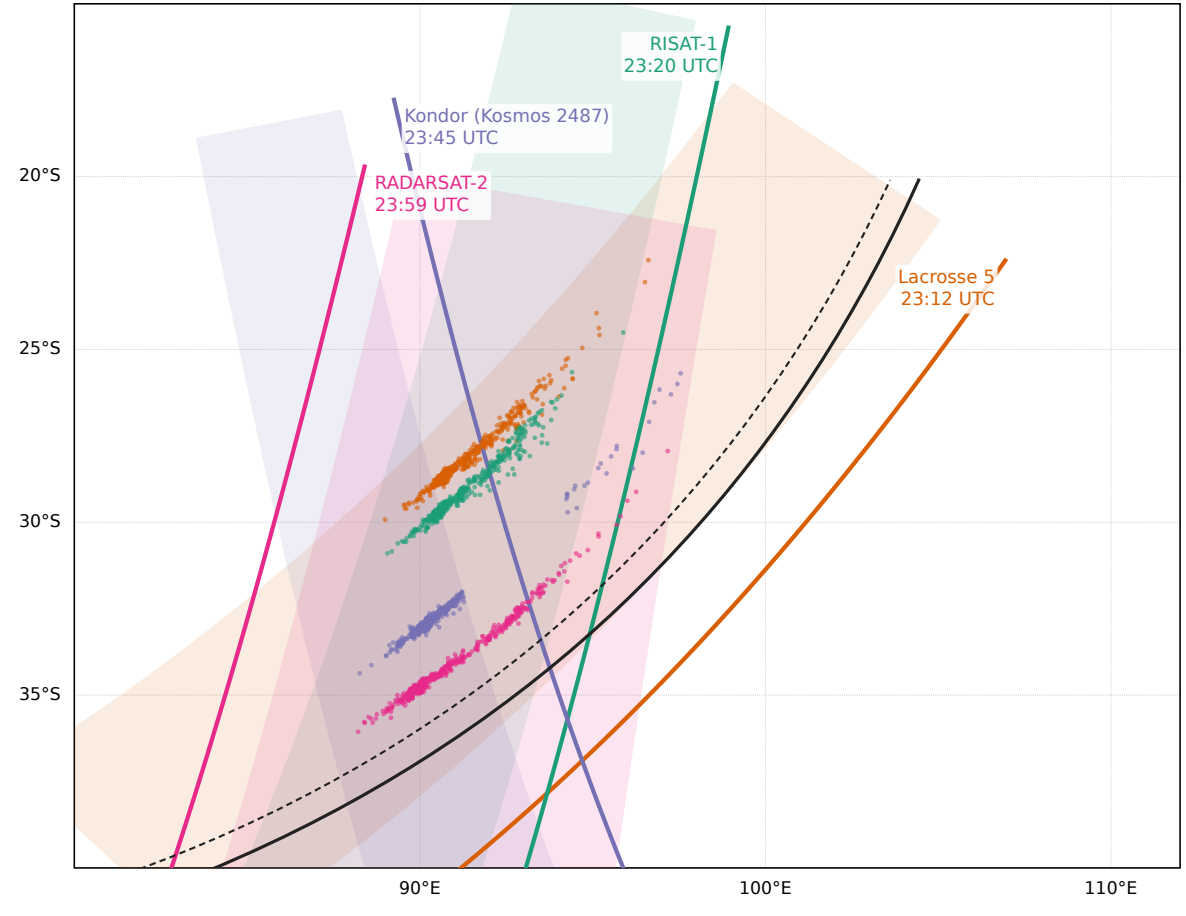
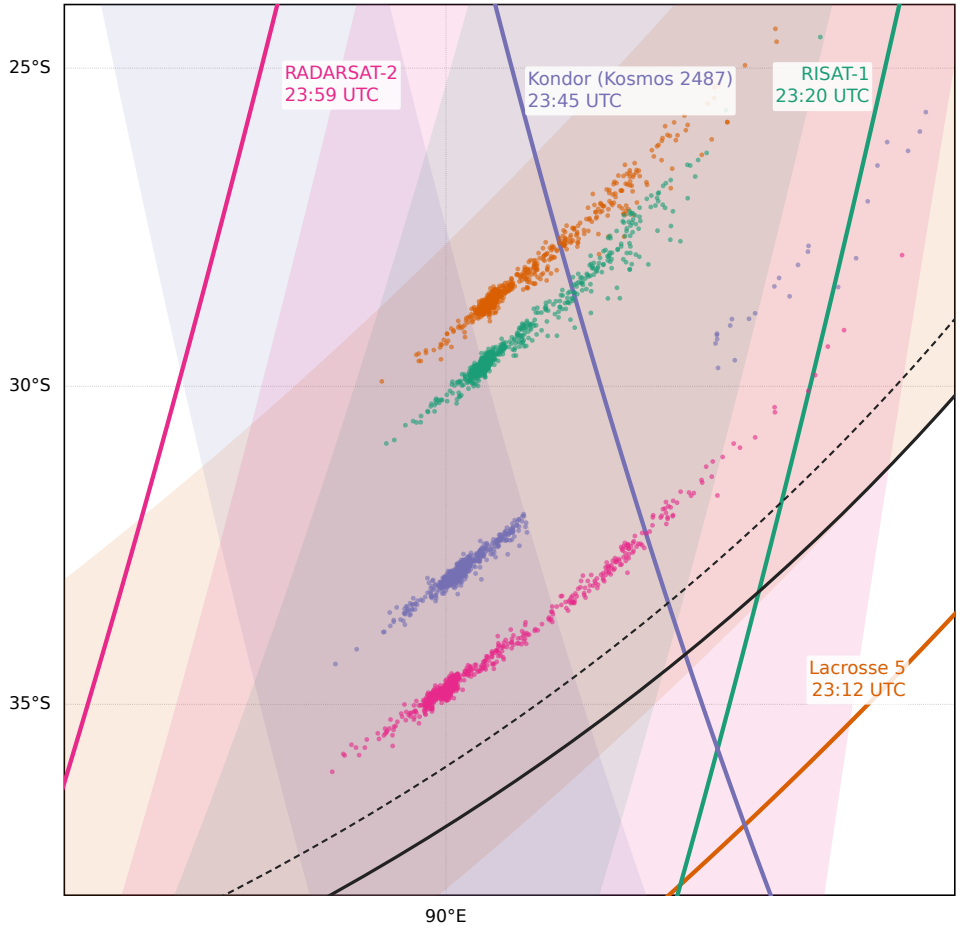
Where four radar satellites could have imaged MH370 in flight, 7–8 March 2014

Coloured line: path of the point below the satellite, 4 min either side of the moment it swept over the aircraft. Tint: the radar's reach on its looking side. Dots: posterior aircraft positions at that moment.
Black solid: 7th arc; dashed: 6th arc.

a Posterior aircraft positions when each radar swept over them, with the radar's reach



b Detail of the southern track (dots: route samples inside each radar's reach)



Plain-language note: each colour is one radar satellite. The dots show where the aircraft probably was when that satellite passed; they lie inside its tinted reach, so it could have imaged the aircraft if it had been ordered to.
Lacrosse 5 (USA) at 23:12, RISAT-1 (India) at 23:20, Kondor, Kosmos 2487 (Russia) at 23:45, RADARSAT-2 (Canada) at 23:59 UTC. Nothing here shows that an image was taken.

Technical: core run J routes.npy, waypoints family (48,000 routes, P(family) 0.9982), linear interpolation along 10-min chords (about 25 NM tolerance at turns); 800 in-reach samples drawn per pass. SGP4 on March 2014 element sets, 5 s steps; reach = ground offsets at the minimum and maximum incidence on the side the aircraft lay (RADARSAT-2 10–60°, RISAT-1 12–55°, grade B; Lacrosse 5 20–60°, Kosmos 2487 20–55°, grade C). Probability inside reach: 1.00, 1.00, 0.80, 0.99. FL350 assumed; the radar would place the aircraft 7–18 km toward itself and 2–18 km along its track from the true position.